

Neuroinflammation as a Shared Mechanism Across Alzheimer's Disease, Parkinson's Disease, and Multiple Sclerosis: A Narrative Review of TNF- α , IL-6, and Microglial Activation

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ABSTRACT

Neuroinflammation has emerged as a candidate shared mechanism linking Alzheimer's disease (AD), Parkinson's disease (PD), and multiple sclerosis (MS), three neurological disorders with distinct clinical presentations but overlapping innate immune features. This narrative review synthesizes published meta-analyses, systematic reviews, positron emission tomography (PET) studies, and postmortem investigations examining tumor necrosis factor-alpha (TNF- α), interleukin-6 (IL-6), and microglial activation markers across the three diseases, with the goal of determining whether the evidence supports a unified inflammatory model or points instead toward disease-specific mechanisms that converge only at the level of downstream effector molecules. Across all three conditions, peripheral and central elevations of TNF- α and IL-6 are consistently reported, and translocator protein (TSPO) positron emission tomography imaging demonstrates increased microglial activation relative to healthy controls. These findings converge on a shared signaling architecture involving toll-like receptor and nuclear factor kappa-B pathways. However, the anatomical distribution, temporal trajectory, and cellular compartmentalization of this activation differ substantially: AD-associated microglial activation concentrates in cortico-limbic and temporo-parietal regions and tracks tau pathology, PD-associated activation is largely restricted to the midbrain, and MS-associated activation is compartmentalized into focal lesions, chronic lesion rims, and diffuse involvement of normal-appearing white matter. Cerebrospinal fluid cytokine profiles are disease-specific rather than uniform, and randomized controlled trials of anti-inflammatory agents have produced markedly different outcomes across the three diseases, with immune-modulating therapies transforming the clinical course of MS while comparable approaches have shown limited efficacy in AD and PD. This review concludes that

neuroinflammation represents a shared downstream effector pathway rather than a single unifying upstream cause, with autoimmune adaptive immunity driving MS and proteinopathy-triggered innate immune activation amplifying neurodegeneration in AD and PD.

Keywords: neuroinflammation; TNF-alpha; interleukin-6; microglial activation; TSPO-PET; Alzheimer's disease; Parkinson's disease; multiple sclerosis

INTRODUCTION

Neuroinflammation, historically viewed as a secondary consequence of neuronal injury, is now understood as an active participant in the pathogenesis of a broad range of central nervous system disorders (1). Alzheimer's disease (AD), Parkinson's disease (PD), and multiple sclerosis (MS) sit at the intersection of this reconceptualization. Although these three diseases differ markedly in their clinical presentation, age of onset, and primary pathological substrate, all three are characterized by activation of the brain's resident innate immune cells, the microglia, alongside elevation of pro-inflammatory cytokines including tumor necrosis factor-alpha (TNF- α) and interleukin-6 (IL-6) (1). This overlap has motivated a growing body of research asking whether AD, PD, and MS share a common inflammatory mechanism that could, in principle, be targeted by a single class of therapeutics, or whether the apparent overlap reflects a shared downstream effector pathway recruited by fundamentally different upstream triggers.

A recent review of the immunology of neurodegenerative disease proposed a useful framework for this question: in some diseases, such as MS, the immune system is the primary driver of pathology, while in others, such as AD and PD, it plays an amplifying role secondary to protein aggregation (2). Testing this framework requires comparing the same biomarkers across all three conditions using comparable methodology, a task complicated by substantial heterogeneity in assay platforms, sample matrices, and imaging protocols across the published literature.

This narrative review synthesizes evidence from peripheral and cerebrospinal fluid (CSF) cytokine studies, translocator protein (TSPO) positron emission tomography (PET) imaging, and postmortem histological investigations of TNF- α , IL-6, and microglial activation across AD, PD, and MS. The review first identifies where findings across the three diseases converge, then examines where they diverge, catalogs major sources of

methodological heterogeneity that complicate cross-disease comparison, and considers what the differential outcomes of anti-inflammatory clinical trials imply about the underlying mechanism. The objective is to evaluate the evidence for and against a unified inflammatory model of neurodegeneration.

DISCUSSION

Convergent Evidence for a Shared Inflammatory Signaling Pathway

The most consistent point of convergence across AD, PD, and MS is the involvement of the same molecular signaling architecture. Toll-like receptors and other pattern-recognition receptors expressed on microglia detect diverse pathological stimuli, amyloid-beta and tau aggregates in AD, aggregated alpha-synuclein in PD, and myelin-derived antigens in MS, and funnel these signals through nuclear factor kappa-B and activator protein-1 pathways to drive production of the same downstream cytokines, including TNF- α , interleukin-1 beta, and IL-6 (1). This shared signaling architecture provides a plausible molecular basis for the repeated observation that TNF- α and IL-6 are elevated in all three diseases, even though the diseases differ in etiology.

In AD, a foundational meta-analysis of 40 studies found significantly elevated peripheral TNF- α and IL-6 in patients compared with controls (3). In PD, a meta-analysis of peripheral blood cytokine data from 13 studies and 898 patients found higher circulating IL-6, TNF, interleukin-1 beta, interleukin-2, interleukin-10, C-reactive protein, and RANTES compared with controls (4), and a more recent systematic review and meta-analysis confirmed elevated peripheral blood and CSF IL-6, TNF- α , and related markers, with parallel elevations observed in CSF (5). In MS, cytokine elevation is documented in stimulated peripheral monocytes and CSF, and a systematic review and meta-analysis of exercise-related cytokine responses in MS patients confirmed baseline elevation of inflammatory markers relative to healthy controls, even though exercise itself did not consistently modulate IL-6 in this population (7).

Microglial activation itself is likewise detectable across all three diseases using both postmortem histology and in vivo TSPO-PET imaging. The first histological demonstration of reactive, HLA-DR-positive microglia in neurodegenerative brain tissue examined both PD substantia nigra and AD cortex in the same study, establishing early evidence that innate immune activation is not unique to one disease (8). A large transdiagnostic meta-analysis of 156 case-control TSPO-PET studies, encompassing 2,381 healthy controls and 2,626 patients

across eleven illness categories, found significantly elevated cortical grey-matter TSPO signal across multiple neurodegenerative and neuroinflammatory conditions, with AD showing among the largest and most widespread cortical elevations of any diagnostic category studied (9).

Divergent Evidence Across Diseases

Despite this convergence at the level of shared signaling molecules, the anatomical distribution, temporal course, and underlying trigger for microglial activation differ substantially across AD, PD, and MS, and these differences bear directly on whether a single unified inflammatory model is adequate to explain the biology.

Anatomical and Temporal Patterns of Microglial Activation

The transdiagnostic TSPO-PET meta-analysis found that cortico-limbic and thalamic elevations in TSPO signal were most prominent for AD, consistent with the disease's characteristic involvement of temporo-parietal and posterior cingulate regions (9). By contrast, TSPO-PET evidence in PD is comparatively restricted; while first-generation TSPO ligands detected diffuse elevation across many brain regions in early studies, this pattern has proven inconsistent with second-generation, more specific ligands, and the most reproducible finding with modern imaging is elevation confined largely to the midbrain (4,8). In MS, TSPO-PET signal is not diffusely elevated across the cortex in the way it is in AD; instead it concentrates in focal white matter lesions, the iron-laden rims of chronic active ("smoldering") lesions, and, to a lesser but still measurable degree, in normal-appearing white matter and deep gray matter structures such as the thalamus (9,10). This anatomical segregation, cortico-limbic in AD, midbrain-restricted in PD, and lesion-compartmentalized in MS, indicates that whatever shared signaling pathway underlies microglial activation in each disease, the pathway is being triggered in anatomically and temporally distinct patterns dictated by where the disease-specific pathological substrate is located.

Autoimmune versus Proteinopathy-Driven Initiation

A second and arguably more fundamental divergence concerns what triggers microglial activation in the first place. In AD and PD, microglia are activated as a response to accumulating misfolded proteins, amyloid-beta and hyperphosphorylated tau in AD, aggregated alpha-synuclein in PD, acting as an innate immune first responder to

endogenous danger signals (1,8). In MS, by contrast, the initiating event is autoimmune: the adaptive immune system, principally T and B lymphocytes, mounts a response against myelin components of the central nervous system, and microglia participate as downstream effectors and antigen-presenting amplifiers of an immune process that originates outside the innate immune system (1,2). This distinction is not merely semantic. A recent review of shared immune targets across neurodegenerative disease explicitly frames MS as a condition in which the immune system is the primary driver of pathology, in contrast to AD, amyotrophic lateral sclerosis, and PD, in which immune activation plays an amplifying rather than initiating role (2). This asymmetry has direct implications for therapeutic strategy, discussed further below.

Compartmentalized versus Diffuse Microglial Activation in Multiple Sclerosis

Postmortem neuropathological work has clarified that microglial activation in MS is not only anatomically distinct from AD and PD but also mechanistically distinct in its relationship to the TSPO-PET signal used to image it. A quantitative neuropathological study comparing TSPO expression across active lesions, chronic active lesion rims, and control white matter found markedly greater TSPO-expressing cell density in active lesions and chronic active lesion rims relative to normal-appearing tissue, and, critically, demonstrated that in human tissue activated microglia do not substantially increase TSPO expression on a per-cell basis; rather, the elevated PET signal in MS lesions reflects an increase in the density of TSPO-expressing cells (10). This finding has broad implications for the interpretation of TSPO-PET data across all three diseases: the assumption, often implicit in the AD and PD imaging literature, that increased TSPO signal directly reflects increased microglial activation per cell may not hold, and cross-disease comparisons of TSPO-PET effect sizes should be interpreted as reflecting differences in microglial density and distribution rather than uniform per-cell activation intensity.

Disease-Specific Cytokine Signatures and Peripheral–CNS Discordance

If neuroinflammation were a single unified process, one would expect the same cytokines to be elevated to a similar degree in the CSF of patients with AD, PD, and MS. This is not what the comparative literature shows. A systematic review and meta-analysis that directly compared CSF cytokine profiles across AD, PD, and amyotrophic lateral sclerosis in a single analytic framework found that AD was characterized by significant elevation of transforming growth factor-beta, monocyte chemoattractant protein-1, and YKL-40, while PD

showed significant elevation of transforming growth factor-beta 1, IL-6, and interleukin-1 beta, a distinguishable, disease-specific profile rather than a single interchangeable inflammatory signature (6). This finding argues against the simplest version of a unified inflammatory model, in which the same cytokines would be expected to serve as universal biomarkers across neurodegenerative disease.

A further complication for cross-disease synthesis is that peripheral (blood) cytokine measurements do not always track central nervous system inflammatory activity. In MS, IL-6 and TNF- α levels in peripheral blood do not reliably move in step with disease activity, and a systematic review and meta-analysis of exercise-related cytokine responses found that IL-6 in particular did not respond to exercise in people with MS the way it characteristically does in healthy individuals, suggesting that peripheral cytokine kinetics in this population are altered in ways not fully explained by CNS inflammatory burden alone (7). This peripheral–CNS discordance means that any meta-analysis pooling blood-based and CSF-based or CNS-imaging-based evidence must treat these as related but non-interchangeable measures of the same underlying process, and it complicates simple claims that peripheral cytokine panels can serve as reliable proxies for the CNS inflammatory state in any of the three diseases.

Methodological Heterogeneity Affecting Cross-Disease Comparison

Beyond the substantive biological divergences described above, several methodological sources of heterogeneity limit the strength of conclusions that can be drawn from cross-disease comparison of the existing literature, and should be explicitly accounted for in any future meta-analysis attempting to quantify the degree of overlap. First, cytokine assay platform varies substantially across studies; enzyme-linked immunosorbent assay, multiplex bead-based panels, and cytometric bead arrays do not produce numerically equivalent results for the same analyte, and this variability has been identified as a substantial contributor to between-study heterogeneity and to publication bias in cytokine meta-analyses of PD (5). Second, sample matrix, serum versus plasma versus CSF, affects absolute cytokine concentrations and their correlation with clinical measures; IL-6 concentrations, for example, are characteristically much lower and more variable in CSF than in serum. Third, for TSPO-PET imaging specifically, the transdiagnostic meta-analysis identified quantification method, whether distribution volume is calculated using an arterial input function versus a reference-tissue approach, as the single largest driver of between-study variance in TSPO-PET effect sizes, explaining approximately one quarter of total variance,

followed by patient age and radioligand generation (9). Because reference-tissue-based studies systematically produce larger effect size estimates than distribution-volume-based studies, cross-disease comparisons that do not stratify by quantification method risk attributing biological differences to what is in fact a methodological artifact. Fourth, the rs6971 polymorphism in the TSPO gene divides individuals into high-, mixed-, and low-affinity binder genotypes for second-generation TSPO radioligands, and inconsistent genotype stratification across the published PD and AD PET literature introduces additional unexplained variance that a rigorous meta-analysis should model as a moderator rather than ignore. Finally, disease stage, duration, and medication status at the time of sampling are inconsistently reported and controlled for across studies, and disease-modifying therapies in MS in particular are known to reduce TSPO-PET signal in white matter lesions, meaning that treatment status is a necessary covariate rather than a nuisance variable in any pooled analysis.

Therapeutic Trial Evidence and Mechanistic Implications

Perhaps the most informative evidence bearing on whether AD, PD, and MS share a truly unified inflammatory mechanism comes not from biomarker studies but from the differential outcomes of clinical trials targeting inflammation in each disease. In MS, immune-modulating disease-modifying therapies, ranging from interferons and glatiramer acetate to more targeted agents such as natalizumab and anti-CD20 monoclonal antibodies, have transformed the clinical trajectory of relapsing forms of the disease, substantially reducing relapse rates and magnetic resonance imaging-detected inflammatory activity. This clinical efficacy is consistent with the framing of MS as a disease in which the immune system is the primary pathological driver (2).

The picture in AD is markedly different. The only randomized, double-blind, placebo-controlled trial of the TNF- α inhibitor etanercept in AD, conducted in 41 participants with mild to moderate disease over a 24-week period, found the drug well tolerated but reported no statistically significant improvement in cognition, behavior, or global function relative to placebo (11). This null result for a targeted anti-TNF agent in AD stands in sharp contrast to the transformative effect of immune therapy in MS, and it is difficult to reconcile with a model in which TNF- α -driven inflammation is a primary causal driver of AD in the same sense that adaptive immune attack on myelin is a primary driver of MS. Minocycline, an antibiotic with anti-inflammatory and

microglia-suppressing properties in preclinical models, has similarly failed to demonstrate clear neuroprotective efficacy in early-phase PD trials.

Taken together, this therapeutic asymmetry, transformative efficacy of immune-targeted therapy in MS versus null or marginal efficacy of comparable approaches in AD and PD, provides some of the strongest available evidence that neuroinflammation, while present and measurable in all three diseases, does not play the same causal role in each. It is consistent with a model in which inflammation is causally upstream and primary in MS, but downstream and amplifying in AD and PD, a distinction that a purely biomarker-based comparison of TNF- α , IL-6, and microglial activation cannot by itself resolve, but that mechanistic and interventional evidence brings into sharper focus (2).

Toward a Unified or Disease-Specific Inflammatory Model

The evidence reviewed here does not support a single, unified inflammatory model in which AD, PD, and MS represent essentially the same underlying process manifesting in different brain regions. The disease-specific CSF cytokine signatures, the anatomically distinct and temporally divergent patterns of microglial activation, the qualitatively different upstream triggers (autoimmune adaptive immunity in MS versus proteinopathy-driven innate immune activation in AD and PD), and the sharply different outcomes of anti-inflammatory clinical trials all argue against treating neuroinflammation as an interchangeable, disease-agnostic mechanism.

At the same time, the evidence equally does not support treating the inflammatory features of these three diseases as coincidental or unrelated. The repeated, cross-disease elevation of TNF- α and IL-6, the shared toll-like receptor and nuclear factor kappa-B signaling architecture, and the consistent detection of activated, TSPO-positive microglia across all three conditions indicate that AD, PD, and MS converge on a shared molecular and cellular effector pathway even as they diverge in what triggers that pathway and in how central its role is to disease causation. The most defensible synthesis of the current evidence is therefore an intermediate model: a shared downstream inflammatory effector system, built on conserved microglial signaling machinery, that is recruited by fundamentally different upstream disease processes and that plays a correspondingly different mechanistic role, primary and causal in MS, amplifying and downstream in AD and PD, in each condition. This model has direct implications for meta-analytic practice and for future research: it suggests that pooling TNF- α , IL-6, and

microglial activation data across AD, PD, and MS without stratifying by disease, and without separately modeling peripheral versus central compartments, risks averaging over biologically meaningful heterogeneity rather than revealing a genuine unifying signal.

CONCLUSION

Across Alzheimer's disease, Parkinson's disease, and multiple sclerosis, $\text{TNF-}\alpha$, IL-6, and microglial activation are reproducibly elevated relative to healthy controls, and this elevation traces back to a shared toll-like receptor and nuclear factor kappa-B signaling architecture in activated microglia. However, the anatomical distribution of this activation, its relationship to disease-specific pathology, the cytokine signatures detected in CSF, and, most tellingly, the outcomes of anti-inflammatory clinical trials differ substantially across the three diseases. These differences indicate that neuroinflammation functions as a shared downstream effector pathway rather than as a single unifying upstream cause of neurodegeneration, with the immune system acting as the primary driver of pathology in MS and as an amplifying, secondary contributor in AD and PD. Future meta-analytic work in this area should treat disease identity as a substantive moderator rather than a nuisance variable, explicitly model the major sources of methodological heterogeneity identified here, including assay platform, sample matrix, TSPO-PET quantification method, and rs6971 genotype, and interpret pooled effect sizes with the understanding that convergent biomarker elevation does not, by itself, establish a convergent causal mechanism.

ACKNOWLEDGMENTS

None.

CONFLICT OF INTEREST

The author declares no conflicts of interest related to this work.

AI DISCLOSURE STATEMENT

During the preparation of this work, the author used Claude (Anthropic) for literature search assistance, synthesis of published meta-analyses and reviews, and drafting support. The author reviewed, verified, and edited the content and takes full responsibility for the accuracy, originality, and integrity of the published work.

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Table 1. Comparison of convergent and divergent neuroinflammatory features across Alzheimer's disease, Parkinson's disease, and multiple sclerosis.

Feature	Alzheimer's Disease	Parkinson's Disease	Multiple Sclerosis
Peripheral TNF- α / IL-6	Elevated (3)	Elevated (4,5)	Elevated (7)
CSF cytokine signature	TGF- β , MCP-1, YKL-40 (6)	TGF- β 1, IL-6, IL-1 β (6)	Elevated IL-6/TNF- α ; activity-linked (7)
Primary microglial trigger	Amyloid- β / tau aggregates	Alpha-synuclein aggregates	Autoimmune anti-myelin response
Anatomical pattern of activation	Cortico-limbic, temporo-parietal, thalamic (9)	Midbrain / nigrostriatal, second-generation ligands (4)	Focal lesions, chronic lesion rims, NAWM, thalamus (9,10)
Compartmentalization	Diffuse; tracks tau spread	Relatively focal (midbrain)	Highly compartmentalized (lesion vs. NAWM vs. GM)
Role of immune system	Amplifying, downstream (2)	Amplifying, downstream (2)	Primary disease driver (2)
Anti-inflammatory trial evidence	Etanercept: no significant benefit (11)	Minocycline: no clear neuroprotection	Immune-modulating DMTs: strong relapse reduction